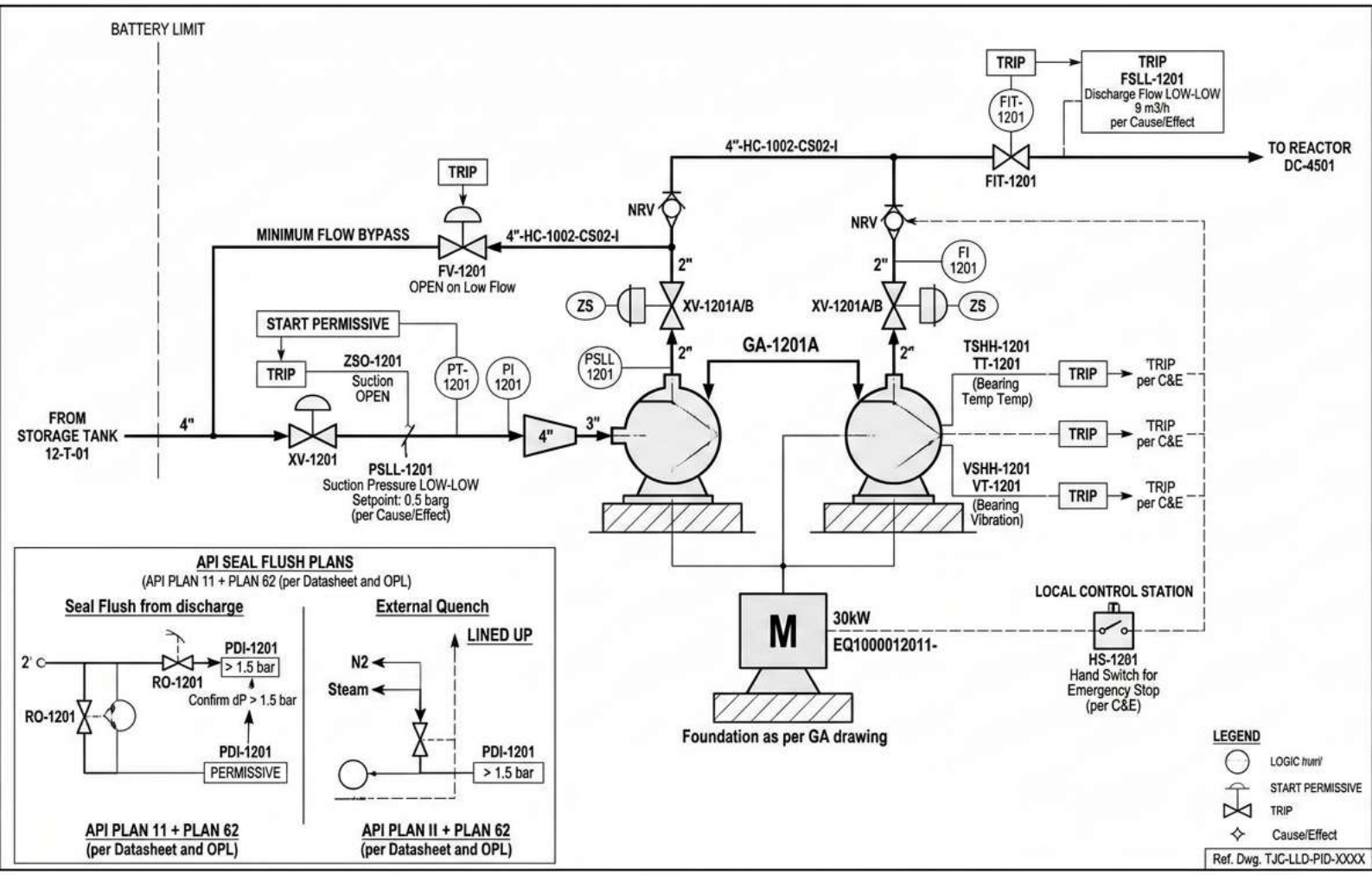
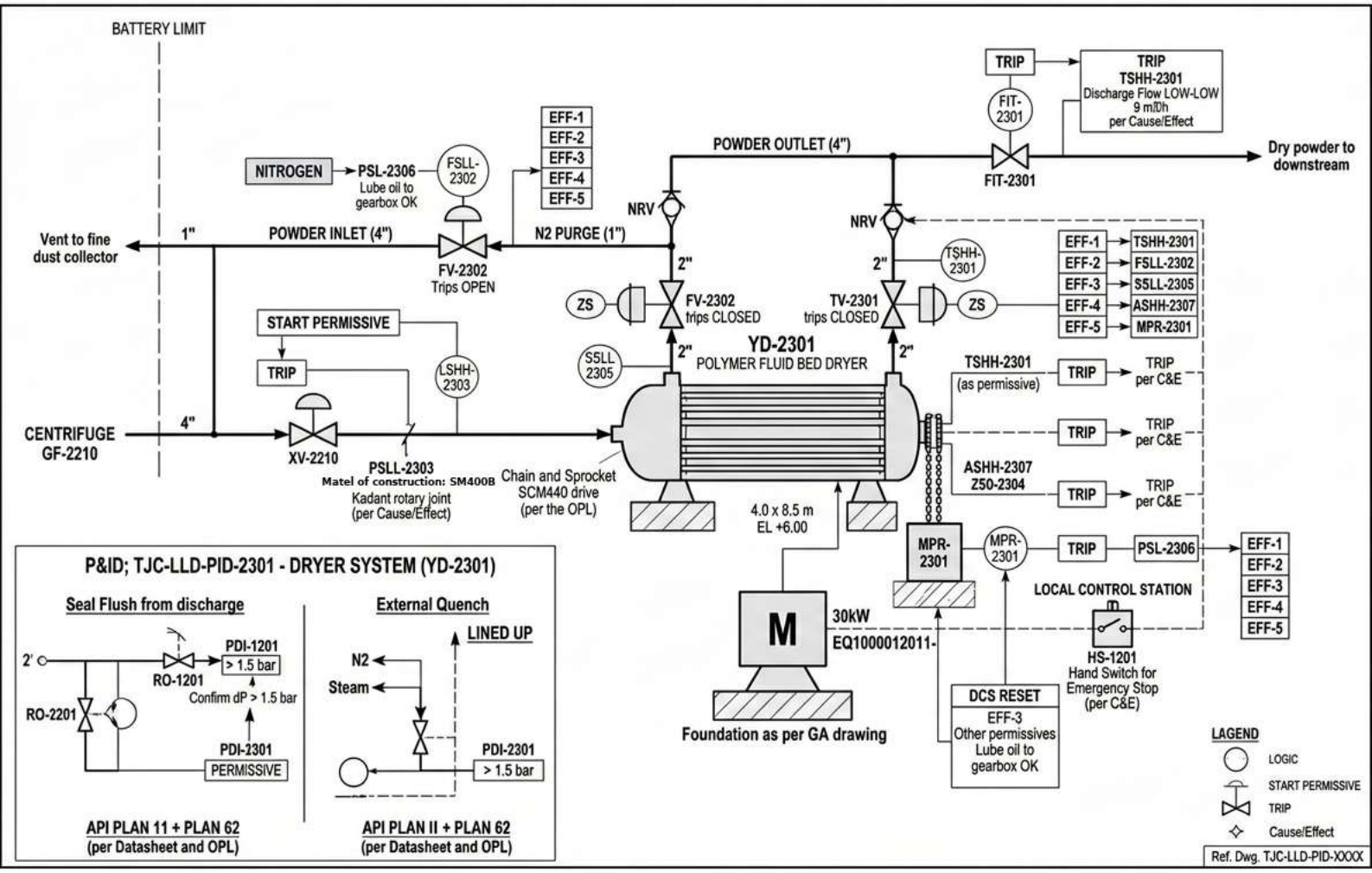


NOTES:

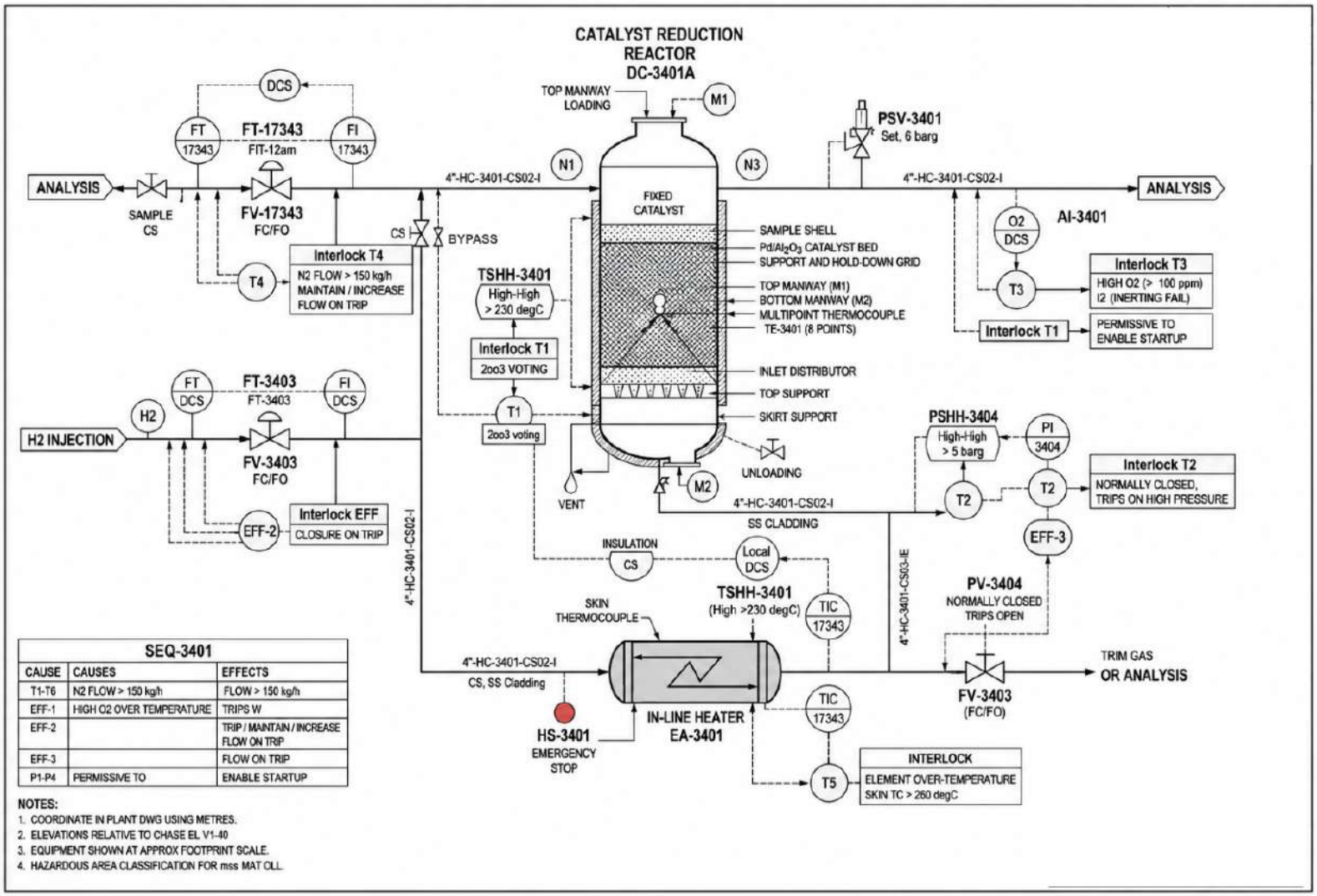
NOTE 1: This is sample data provided for CALIBER purposes only;

NOTE 2: Refer to Datasheet TJC-LLD-DS-GA-1201A for design parameters;

NOTE 3: See SEQ-1201 logic diagram.

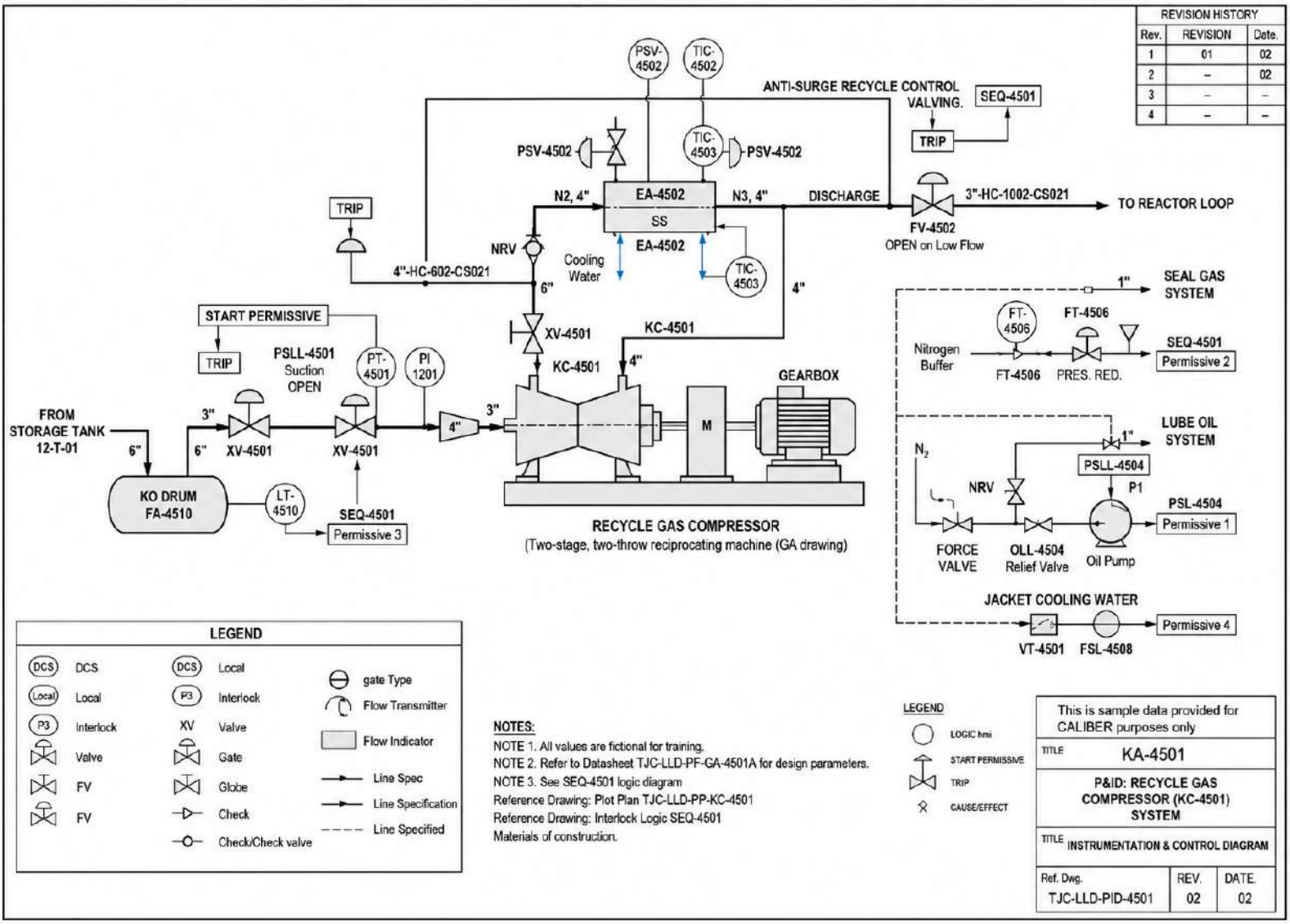


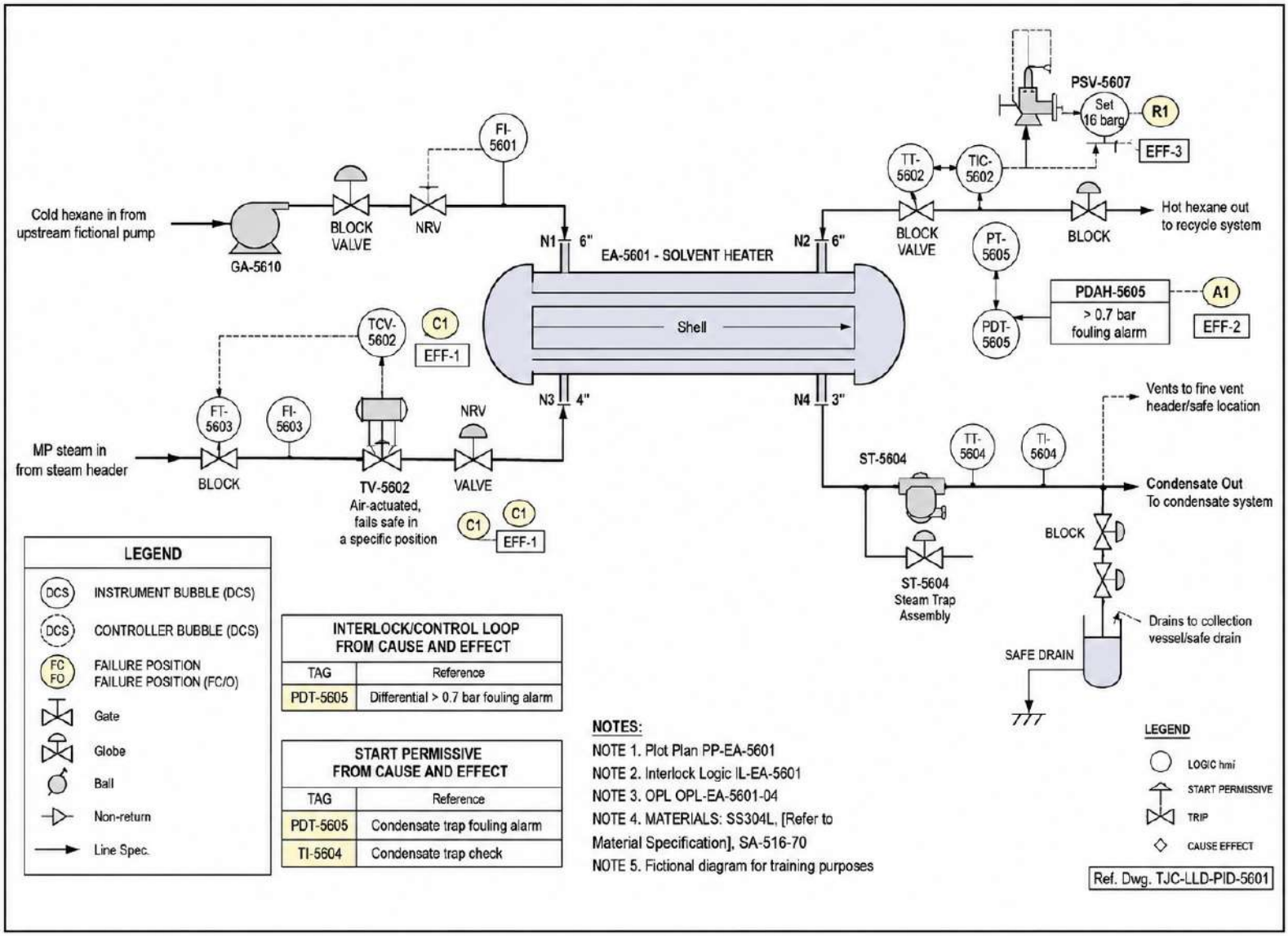
NOTE 1: This is sample data provided for CALIBER purposes only.
NOTE 2: Refer to Datasheet TJC-LLD-DS-GA-1201A for design parameters.
NOTE 3: See SEQ-1201 logic diagram

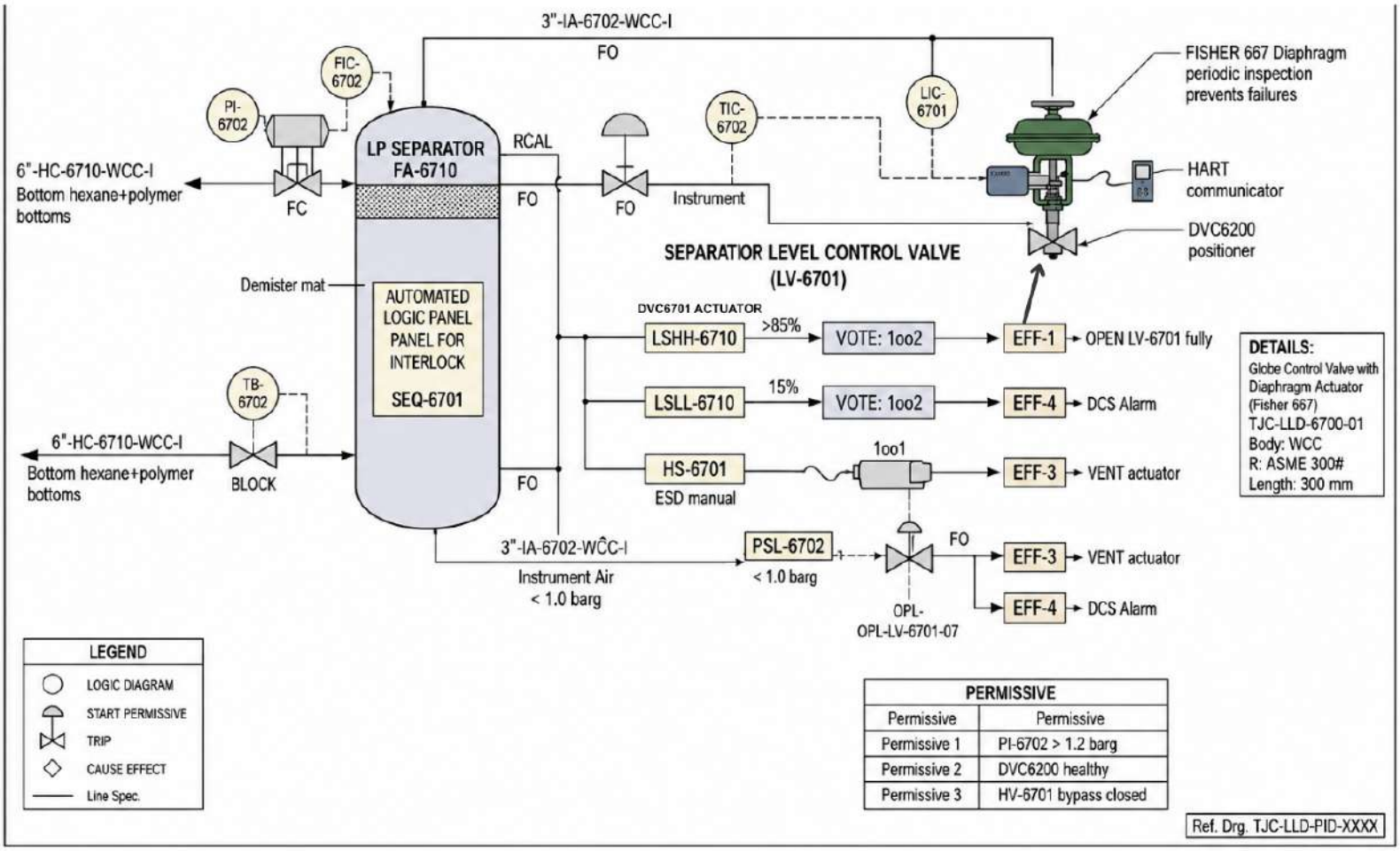


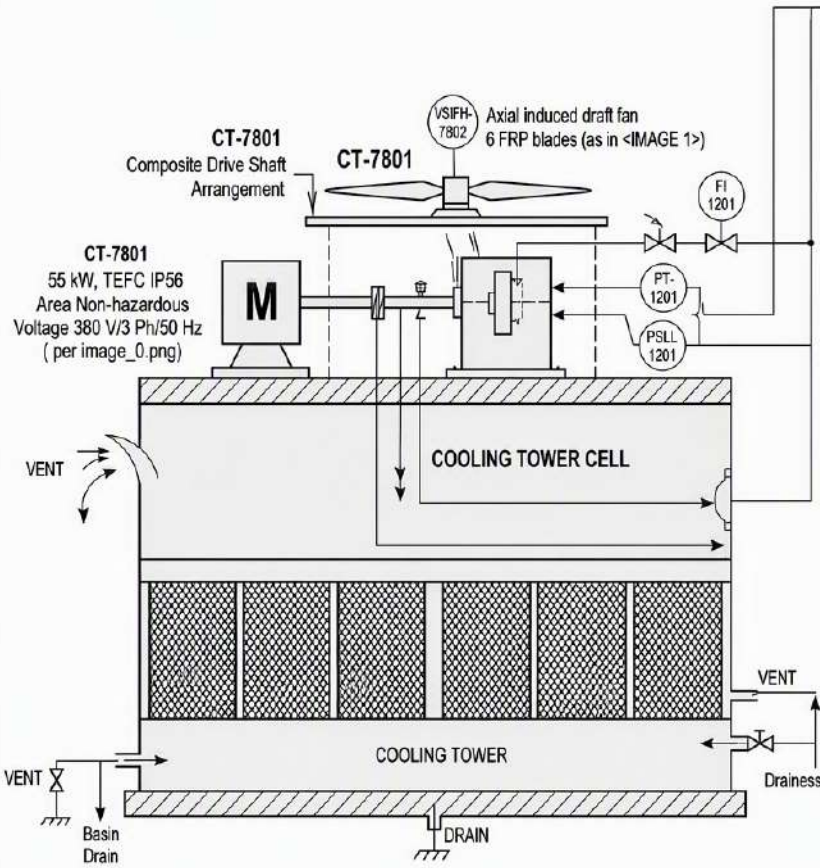
SEQ-3401		
CAUSE	CAUSES	EFFECTS
T1-T6	N2 FLOW > 150 kg/h	FLOW > 150 kg/h
EFF-1	HIGH O2 OVER TEMPERATURE	TRIPS W
EFF-2		TRIP / MAINTAIN / INCREASE FLOW ON TRIP
EFF-3		FLOW ON TRIP
P1-P4	PERMISSIVE TO	ENABLE STARTUP

NOTES:
1. COORDINATE IN PLANT DWG USING METRES.
2. ELEVATIONS RELATIVE TO CHASE EL V1-40
3. EQUIPMENT SHOWN AT APPROX FOOTPRINT SCALE.
4. HAZARDOUS AREA CLASSIFICATION FOR mss MAT CLL.


REVISION HISTORY
Rev. | REVISION | Date.
1 | 01 | 02
2 | – | 02
3 | – | –
4 | – | –
PSV-4502
TIC-4502
ANTI-SURGE RECYCLE CONTROL VALVING.
SEQ-4501
TRIP
PSV-4502
TIC-4503
PSV-4502
N2, 4"
EA-4502
SS
N3, 4"
DISCHARGE
3"-HC-1002-CS021
TO REACTOR LOOP
FV-4502
OPEN on Low Flow
TRIP
NRV
Cooling Water
EA-4502
TIC-4503
4"-HC-602-CS021
6"
4"
SEAL GAS SYSTEM
1"
FT-4506
FT-4506
Nitrogen Buffer
FT-4506
PRES. RED.
SEQ-4501
Permissive 2
START PERMISSIVE
TRIP
PSLL-4501
Suction OPEN
PT-4501
PI 1201
XV-4501
KC-4501
KC-4501
4"
GEARBOX
FROM STORAGE TANK 12-T-01
3"
6"
6"
XV-4501
XV-4501
4"
3"
M
LUBE OIL SYSTEM
1"
N2
PSLL-4504
NRV
P1
PSL-4504
Permissive 1
FORCE VALVE
OLL-4504 Relief Valve
Oil Pump
KO DRUM FA-4510
LT-4510
SEQ-4501
Permissive 3
RECYCLE GAS COMPRESSOR
(Two-stage, two-throw reciprocating machine (GA drawing)
JACKET COOLING WATER
Permissive 4
VT-4501
FSL-4508
LEGEND
DCS DCS
Local Local
P3 Interlock
Valve
FV
FV
DCS Local
P3 Interlock
XV Valve
Gate
Globe
Check
Check/Check valve
gate Type
Flow Transmitter
Flow Indicator
Line Spec
Line Specification
Line Specified
NOTES:
NOTE 1. All values are fictional for training.
NOTE 2. Refer to Datasheet TJC-LLD-PF-GA-4501A for design parameters.
NOTE 3. See SEQ-4501 logic diagram
Reference Drawing: Plot Plan TJC-LLD-PP-KC-4501
Reference Drawing: Interlock Logic SEQ-4501
Materials of construction.
LEGEND
LOGIC hmi
START PERMISSIVE
TRIP
CAUSE/EFFECT
This is sample data provided for CALIBER purposes only
TITLE KA-4501
P&ID: RECYCLE GAS COMPRESSOR (KC-4501) SYSTEM
TITLE INSTRUMENTATION & CONTROL DIAGRAM
Ref. Dwg. TJC-LLD-PID-4501
REV. 02
DATE. 02


Cold hexane in from upstream fictional pump
GA-5610
BLOCK VALVE
NRV
FI-5601
N1 6"
EA-5601 - SOLVENT HEATER
Shell
N2 6"
N3 4"
N4 3"
PSV-5607
Set 16 barg
R1
EFF-3
TT-5602
TIC-5602
BLOCK VALVE
BLOCK
Hot hexane out to recycle system
PT-5605
PDT-5605
PDAH-5605
> 0.7 bar fouling alarm
A1
EFF-2
TCV-5602
C1
EFF-1
FT-5603
FI-5603
MP steam in from steam header
BLOCK
TV-5602
Air-actuated, fails safe in a specific position
NRV
VALVE
C1
EFF-1
Vents to fine vent header/safe location
TT-5604
TI-5604
ST-5604
Condensate Out To condensate system
BLOCK
ST-5604 Steam Trap Assembly
Drains to collection vessel/safe drain
SAFE DRAIN
LEGEND
DCS INSTRUMENT BUBBLE (DCS)
DCS CONTROLLER BUBBLE (DCS)
FC FO FAILURE POSITION FAILURE POSITION (FC/O)
Gate
Globe
Ball
Non-return
Line Spec.
INTERLOCK/CONTROL LOOP FROM CAUSE AND EFFECT
TAG | Reference
PDT-5605 | Differential > 0.7 bar fouling alarm
START PERMISSIVE FROM CAUSE AND EFFECT
TAG | Reference
PDT-5605 | Condensate trap fouling alarm
TI-5604 | Condensate trap check
NOTES:
NOTE 1. Plot Plan PP-EA-5601
NOTE 2. Interlock Logic IL-EA-5601
NOTE 3. OPL OPL-EA-5601-04
NOTE 4. MATERIALS: SS304L, [Refer to Material Specification], SA-516-70
NOTE 5. Fictional diagram for training purposes
LEGEND
LOGIC hmi
START PERMISSIVE
TRIP
CAUSE EFFECT
Ref. Dwg. TJC-LLD-PID-5601


3"-IA-6702-WCC-I
FO
FIC-6702
PI-6702
LP SEPARATOR FA-6710
RCAL
FO
FO
TIC-6702
LIC-6701
FISHER 667 Diaphragm periodic inspection prevents failures
HART communicator
DVC6200 positioner
6"-HC-6710-WCC-I
Bottom hexane+polymer bottoms
FC
Instrument
SEPARATOR LEVEL CONTROL VALVE (LV-6701)
Demister mat
AUTOMATED LOGIC PANEL PANEL FOR INTERLOCK
SEQ-6701
DVC6701 ACTUATOR
LSHH-6710
>85%
VOTE: 1oo2
EFF-1
OPEN LV-6701 fully
LSLL-6710
15%
VOTE: 1oo2
EFF-4
DCS Alarm
TB-6702
6"-HC-6710-WCC-I
Bottom hexane+polymer bottoms
BLOCK
HS-6701
ESD manual
1oo1
EFF-3
VENT actuator
FO
3"-IA-6702-WCC-I
Instrument Air
< 1.0 barg
PSL-6702
< 1.0 barg
FO
EFF-3
VENT actuator
EFF-4
DCS Alarm
OPL-
OPL-LV-6701-07
DETAILS:
Globe Control Valve with Diaphragm Actuator (Fisher 667)
TJC-LLD-6700-01
Body: WCC
R: ASME 300#
Length: 300 mm
LEGEND
LOGIC DIAGRAM
START PERMISSIVE
TRIP
CAUSE EFFECT
Line Spec.
PERMISSIVE
Permissive | Permissive
Permissive 1 | PI-6702 > 1.2 barg
Permissive 2 | DVC6200 healthy
Permissive 3 | HV-6701 bypass closed
Ref. Drg. TJC-LLD-PID-XXXX



INTERLOCK SEQ-7801

IDlimitator	Tag	Test/ act.	Effects EFF-1		EFF-2		EFF-3		Final Element / Action
Fan vibration	VSHH-7802	>9 mm/s	1	X	T1	T2	T3	T4	Trip Fan Motor CT-7801
Fan vibration	VSHH-7802	=90 dag C	1	X		X		X	Trip Fan Motor CT-7801
Gearbox oil temp	TSHH-7802	=90 dag C	1			X	X		Annunciate Alarm on DCS
Gearbox oil pressure	PSL-7807	0.8 barg	1	X		X			Request Start of Spare Coil CT-7802
Fan motor overload	MPR-7801	MPR pickup	1					X	Request Start of Spare Coil CT-7802
								→	Spare Coil P&ID
									START PERMISSIVE (AND-gate)
									- LG Gearbox oil level OK - LT-7804 Basin level normal Basin level normal (LSL-7804) - lockout release logic - L Gearbox oil level OK < Other (LSL-7804) - Lockout release logic in orIf per image_0.png

LEGEND

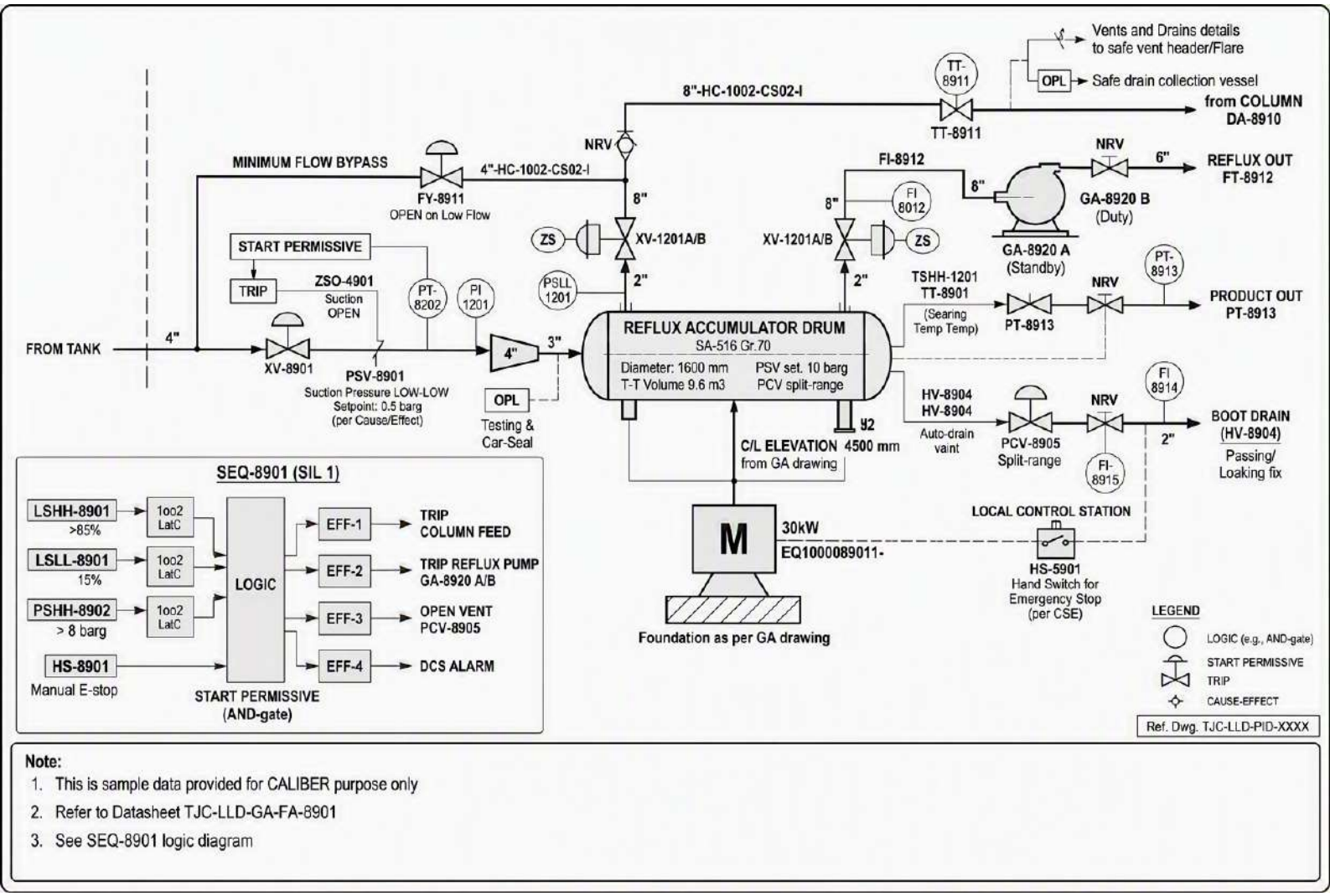
- LOGIC hum/
- START PERMISSIVE
- TRIP
- CAUSE EFFECT

Ref. Plog. TJC-LLD-PID-XXXX

P&ID: COOLING TOWER FAN (CT-7801) SYSTEM

NOTES:

NOTE 1. This is sample data provided for CALIBER purpose only;
NOTE 2: Refer to Datasheet TJC-LLD-DS-GA-7801A for design parameters;
NOTE 3: See SEQ-7801 logic diagram.



Note:

1. This is sample data provided for CALIBER purpose only
2. Refer to Datasheet TJC-LLD-GA-FA-8901
3. See SEQ-8901 logic diagram